NATIONAL UNIVERSITY OF SINGAPORE

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MA1522 Linear Algebra for Computing **Graded Quiz 4**

1. Let $\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n$ be distinct vectors in $\mathbb{R}^n$.

(1) If $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ is linearly independent, then it is a basis for $\mathbb{R}^n$.

(2) If $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ spans $\mathbb{R}^n$, then it is a basis for $\mathbb{R}^n$.

Solution. Note that $\dim(\mathbb{R}^n) = n$.

(1) True; see Pg 108 of Chapter 3.

(2) True; see Pg 108 of Chapter 3.

2. Let V be a subspace of $\mathbb{R}^n$, and let S be a basis for V. Let $T = \{\boldsymbol{v}_1,\boldsymbol{v}_2,\ldots,\boldsymbol{v}_k\} \subseteq V$, and let $\boldsymbol{A} = \begin{pmatrix}[\boldsymbol{v}_1]_S & [\boldsymbol{v}_2]_S & \cdots & [\boldsymbol{v}_k]_S\end{pmatrix}$. Let $\boldsymbol{R}$ be a row-echelon form of $\boldsymbol{A}$.

(1) $\operatorname{span}(T) = V$ if and only if all rows of $\boldsymbol{R}$ are nonzero.

(2) T is linearly independent if and only if all columns of $\boldsymbol{R}$ are pivot.

Solution. Let $\dim(V) = |S| = m$.

(1) True. $\operatorname{span}(T) = V \Leftrightarrow \operatorname{span}\{[\boldsymbol{v}_1]_S,[\boldsymbol{v}_2]_S,\ldots,[\boldsymbol{v}_k]_S\} = \mathbb{R}^m$ (Pg 157 of Chapter 3) $\Leftrightarrow$ all rows of $\boldsymbol{R}$ are nonzero (Pg 37 of Chapter 3).

(2) True. T is linearly independent $\Leftrightarrow \{[\boldsymbol{v}_1]_S,[\boldsymbol{v}_2]_S,\ldots,[\boldsymbol{v}_k]_S\}$ is linearly independent (Pg 157 of Chapter 3) $\Leftrightarrow$ all columns of $\boldsymbol{R}$ are pivot (Pg 85 of Chapter 3).

3. Let S be a basis for $\mathbb{R}^n$, and let $\boldsymbol{P}$ be a square matrix of order n.

(1) If $\boldsymbol{P}$ is the transition matrix from S to some basis T for $\mathbb{R}^n$, then $\boldsymbol{P}$ is invertible.

(2) If $\boldsymbol{P}$ is invertible, then $\boldsymbol{P}$ is the transition matrix from S to some basis T for $\mathbb{R}^n$.

Solution. (1) True; implicit in Pg 153 of Chapter 3.

(2) True. If $\boldsymbol{P}$ is the transition matrix from $S = \{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\}$ to a basis $T = \{\boldsymbol{w}_1,\ldots,\boldsymbol{w}_n\}$, then the reduced row-echelon form of $(\boldsymbol{A} \mid \boldsymbol{B})$ is $(\boldsymbol{I} \mid \boldsymbol{P})$, where $\boldsymbol{A} = \begin{pmatrix}\boldsymbol{w}_1 & \cdots & \boldsymbol{w}_n\end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix}\boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n\end{pmatrix}$. But the reduced row-echelon form of $(\boldsymbol{A} \mid \boldsymbol{B})$ equals $(\boldsymbol{I} \mid \boldsymbol{A}^{-1}\boldsymbol{B})$. Thus $\boldsymbol{P} = \boldsymbol{A}^{-1}\boldsymbol{B}$, and hence $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{P}^{-1}$. This shows that $\boldsymbol{P}$ is the transition matrix from S to T when we define T to be the columns of $\boldsymbol{B}\boldsymbol{P}^{-1}$.

4. Let $\boldsymbol{A}$ be a matrix and let $\boldsymbol{R}$ be a row-echelon form of $\boldsymbol{A}$.

(1) The nullspace of $\boldsymbol{A}$ equals the nullspace of $\boldsymbol{R}$.

(2) The row space of $\boldsymbol{A}$ equals the row space of $\boldsymbol{R}$.

(3) The column space of $\boldsymbol{A}$ equals the column space of $\boldsymbol{R}$.

Solution. (1) True; the solution set of $\boldsymbol{Ax} = \boldsymbol{0}$ equals the solution set of $\boldsymbol{Rx} = \boldsymbol{0}$.

(2) True; see Pg 9 of Chapter 4.

(3) False. Let $\boldsymbol{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then $\boldsymbol{R} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is a row-echelon form of $\boldsymbol{A}$. Clearly

$$\text{column space of } \boldsymbol{A} = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} \,\middle|\, x \in \mathbb{R} \right\} \neq \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} \,\middle|\, x \in \mathbb{R} \right\} = \text{column space of } \boldsymbol{R}.$$

5. Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $\boldsymbol{B}$ be an $n \times m$ matrix. Suppose that $\boldsymbol{AB} = \boldsymbol{I}$.

(1) $\operatorname{rank}(\boldsymbol{A}) = m$.

(2) $\operatorname{rank}(\boldsymbol{B}) = n$.

(3) $\boldsymbol{BA} = \boldsymbol{I}$.

Solution. (1) True; see Pg 49 of Chapter 4.

(2) False. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $\boldsymbol{AB} = (1)$ and $\operatorname{rank}(\boldsymbol{B}) = 1 \neq 2$.

(3) False. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $\boldsymbol{AB} = (1)$ and $\boldsymbol{BA} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \boldsymbol{I}$.

6. Let $\boldsymbol{A}$ be a matrix and let $\boldsymbol{R}$ be a row-echelon form of $\boldsymbol{A}$.

(1) The rows of $\boldsymbol{R}$ containing the pivot points form a basis for the row space of $\boldsymbol{A}$.

(2) The columns of $\boldsymbol{R}$ containing the pivot points form a basis for the column space of $\boldsymbol{A}$.

Solution. (1) True. The nonzero rows of $\boldsymbol{R}$ form basis for the row space of $\boldsymbol{A}$, the a row of $\boldsymbol{R}$ is nonzero if and only if it contains a pivot point.

(2) False. Let $\boldsymbol{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Then $\boldsymbol{R} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is a row-echelon form of $\boldsymbol{A}$. Since the column space of $\boldsymbol{A}$ and the column space of $\boldsymbol{R}$ are not the same, the columns of $\boldsymbol{R}$ cannot span the column space of $\boldsymbol{A}$, and hence cannot be a basis for the latter.

7. Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $\boldsymbol{B}$ be an $n \times p$ matrix.

(1) $\operatorname{rank}(\boldsymbol{AB}) \leq \min\{\operatorname{rank}(\boldsymbol{A}), \operatorname{rank}(\boldsymbol{B})\}$.

(2) $\operatorname{rank}(\boldsymbol{AB}) \leq \max\{\operatorname{rank}(\boldsymbol{A}), \operatorname{rank}(\boldsymbol{B})\}$.

Solution. (1) True; see Pg 35 of Chapter 4.

(2) True; $\operatorname{rank}(\boldsymbol{AB}) \leq \min\{\operatorname{rank}(\boldsymbol{A}), \operatorname{rank}(\boldsymbol{B})\} \leq \max\{\operatorname{rank}(\boldsymbol{A}), \operatorname{rank}(\boldsymbol{B})\}$.

8. Let $\boldsymbol{A}$ be an $m \times n$ matrix, let $\boldsymbol{B}$ be an $n \times p$ matrix and let $\boldsymbol{C}$ be an invertible matrix of order n.

(1) $\operatorname{rank}(\boldsymbol{AC}) = \operatorname{rank}(\boldsymbol{A})$.

(2) $\operatorname{rank}(\boldsymbol{CB}) = \operatorname{rank}(\boldsymbol{B})$.

Solution. (1) True. Since $\boldsymbol{C}$ is invertible, column space of $\boldsymbol{C} = \{\boldsymbol{Cv} \mid \boldsymbol{v} \in \mathbb{R}^n\} = \mathbb{R}^n$, so that

$$\begin{aligned}\text{column space of } \boldsymbol{AC} &= \{\boldsymbol{ACu} \mid \boldsymbol{u} \in \mathbb{R}^n\} \\ &= \{\boldsymbol{Av} \mid \boldsymbol{v} = \boldsymbol{Cu} \text{ for some } \boldsymbol{u} \in \mathbb{R}^n\} \\ &= \{\boldsymbol{Av} \mid \boldsymbol{v} \in \mathbb{R}^n\} = \text{column space of } \boldsymbol{A}.\end{aligned}$$

Taking dimensions then yields the required equality.

(2) True. Since $\boldsymbol{C}$ is invertible, $\boldsymbol{C}^{\mathrm{T}}$ is also invertible. Thus

$$\operatorname{rank}(\boldsymbol{CB}) = \operatorname{rank}((\boldsymbol{CB})^{\mathrm{T}}) = \operatorname{rank}(\boldsymbol{B}^{\mathrm{T}}\boldsymbol{C}^{\mathrm{T}}) = \operatorname{rank}(\boldsymbol{B}^{\mathrm{T}}) = \operatorname{rank}(\boldsymbol{B}).$$

9. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n.

(1) The row space of $\boldsymbol{AB}$ is a subspace of the row space of $\boldsymbol{B}$.

(2) The column space of $\boldsymbol{AB}$ is a subspace of the column space of $\boldsymbol{B}$.

Solution. (1) True; see Slide 30 of Chapter 4.

(2) False. Let $\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $\boldsymbol{AB} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.
Thus column space of $\boldsymbol{AB} = \{(0,x)^{\mathrm{T}} \mid x \in \mathbb{R}\} \not\subseteq \{(x,0)^{\mathrm{T}} \mid x \in \mathbb{R}\} =$ column space of $\boldsymbol{B}$.

10. Let $\boldsymbol{A}$ be a square matrix of order n.

(1) $\boldsymbol{A}$ is invertible if and only if $\operatorname{rank}(\boldsymbol{A}) = n$.

(2) $\boldsymbol{A}$ is invertible if and only if $\operatorname{nullity}(\boldsymbol{A}) = 0$.

Solution. (1) True. $\boldsymbol{A}$ is invertible $\Leftrightarrow$ the columns of $\boldsymbol{A}$ form a basis for $\mathbb{R}^n \Leftrightarrow \operatorname{rank}(\boldsymbol{A}) = n$.

(2) True. $\boldsymbol{A}$ is invertible $\Leftrightarrow \boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution $\Leftrightarrow \operatorname{Null}(\boldsymbol{A}) = \{\boldsymbol{0}\} \Leftrightarrow \operatorname{nullity}(\boldsymbol{A}) = 0$.